ETHAN LUH

COMPUTER SCIENCE • MATHEMATICS

CONTACT

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EDUCATION

B.S. Computer Science
& Mathematics

Penn State Behrend

Expected May 2028

SKILLS

Languages

- Python
- C++
- C
- LaTeX

Areas

- Graph algorithms
- Machine learning
- NLP
- Statistical modeling
- Lattice polytopes
- Combinatorics

Tools

- Git
- NumPy
- PyTorch
- scikit-learn
- OpenCV

ACTIVITIES

President, Volleyball Club
Penn State Behrend • 2024–Present

Peer Tutor
CS & Math • 2023–Present

Math Tutor
Mathnasium • 2021–2023

PROFILE

CS/Math double major pursuing a PhD in CS Theory. Research in Ehrhart theory and lattice-polytope combinatorics under Dr. Derek Hanely; builds software that spans theoretical enumeration and production engineering.

EXPERIENCE

Software Engineering Intern — Summer 2026
AgentMade

- Built and open-sourced Quire (see Projects), a triage multiplexer for AI-generated pull requests, as the core engineering contribution of the internship.

Undergraduate Researcher — 2024–Present
Penn State Behrend — Advisor: Dr. Derek Hanely

- **Applying Ehrhart theory** (IDP, h^*-vector unimodality, unimodular triangulations) to zero forcing polytopes on small graph families.
- **Computational enumeration** via Python; investigating connections between lattice polytope geometry and graph-theoretic parameters.

PROJECTS

Quire — Python, TypeScript • Open Source
Triage multiplexer for AI-generated pull requests

- **Groups PRs by declared product direction** into bundles, cutting reviewer decision load from one-per-PR to one-per-bundle.
- **Multi-tenant architecture** with fully isolated per-team state and disk-based JSON persistence in place of a database.
- **Direction-drift detection**: flags PRs whose changes silently exceed their declared scope.

PreferenceLayer — Python, FastAPI
Portable preference credentials + server-side quality intelligence for agentic shopping

- **PTP: user-owned credentials** — Ed25519-signed W3C Verifiable-Credential envelope with local differential-privacy updates and an OAuth device flow for cross-platform sync.
- **QIL: Bayesian quality aggregation** — hierarchical Beta-Binomial aggregation over product failure rates plus Gaussian Process modeling for continuous dimensions, served via FastAPI and MCP tool bindings.
- **Preference/quality blending**: learns an α-weighted combination of PTP credentials and QIL posteriors into item scores, evaluated via NDCG/transfer experiments.

market-sentiment-predictor — Python • Open Source
News-driven quantile stock-return prediction pipeline

- **Multi-source ingestion & FinBERT sentiment**: fuses price/volume, news, SEC filings, and Reddit signals into continuous sentiment scores weighted by source credibility and recency decay.
- **Graph diffusion for impact timing**: heat kernel over a directed actor-category graph (SEC/Corp $\to$ Institutional $\to$ Press $\to$ Retail) estimates the lag before price impact, with edges calibrated from historical events.
- **Quantile regression & backtesting**: predicts P10/P50/P90 log-returns across horizons conditioned on VIX regime; benchmarked against momentum/ARIMA/GARCH baselines in a walk-forward backtest, served via FastAPI.